Codebook — Hands-on Teaching Data

GLSS6 2012/13 → DHS 2014 synthetic-data practical | IndabaX Ghana 2026 (ACET)

This codebook describes every variable in the prepared teaching data for the hands-on session. **Two files:** a *household* file (one row per household; GLSS and DHS stacked) and a *child* file (one row per under-five child, linked to a DHS household by `hh_key`). GLSS supplies welfare; DHS supplies child outcomes; shared predictors bridge the two. "Source = both" means the variable is observed and harmonized in both surveys.

Household file — shared predictors (X)

Variable	Source	Type	Description	Values / units
`region10`	both	categorical	Administrative region (10-region frame)	1 Western, 2 Central, 3 Greater Accra, 4 Volta, 5 Eastern, 6 Ashanti, 7 Brong Ahafo, 8 Northern, 9 Upper East, 10 Upper West
`urban`	both	binary	Urban (vs rural) location	1 = urban, 0 = rural
`hh_size`	both	count	Number of household members	integer
`members_total`	both	count	Household members from the roster	integer (approx hh_size)
`female_members`	both	count	Female household members	integer
`children_under5`	both	count	Members under age 5	integer
`children_under15`	both	count	Members under age 15	integer
`adults_15plus`	both	count	Members aged 15 or older	integer
`elders_65plus`	both	count	Members aged 65 or older	integer
`head_age`	both	count	Age of household head (years)	integer
`head_female`	both	binary	Household head is female	1 = yes, 0 = no
`head_educ_cat`	both	categorical	Education of head (harmonized)	none_or_primary, basic_or_middle, secondary_or_vocational, higher
`has_electricity`	both	binary	Household has electricity	1 = yes, 0 = no
`water_source_cat`	both	categorical	Main drinking-water source	piped_dwelling_or_yard, public_or_neighbor_piped, borehole_tube_well, protected_well_spring, surface_or_unprotected, rain_tanker_vendor, packaged_water, other
`toilet_type_screen`	both	categorical	Toilet facility (public toilet folded into pit/KVIP)	flush_or_wc, pit_or_kvip, no_facility, bucket_hanging_other
`cooking_fuel_cat`	both	categorical	Main cooking fuel	gas_or_electricity, kerosene, charcoal, wood, other_biomass_or_solid, no_cooking
`floor_material_screen`	both	categorical	Floor material (finished/tiled folded into cement)	earth_mud_dung, cement_or_finished, wood_brick_stone, other
`floor_non_earth`	both	binary	Floor is not bare earth/mud	1 = non-earth, 0 = earth
`roof_material_cat`	both	categorical	Roof material	natural_thatch_bamboo, sheet_metal_asbestos_slate, cement_tile_shingle, wood, other
`sleeping_rooms`	both	count	Rooms used for sleeping	integer

Household file — synthetic targets (Y, GLSS-only) and IDs

The targets below are observed in **GLSS only**. In DHS they are blank — they are what the practical *generates*. `poverty_status` is **derived** from `welfare`, never modeled directly.

Variable	Source	Type	Description	Values / units
`welfare`	GLSS	continuous	Consumption-based welfare per adult equivalent (main synthetic target)	GHS per year (2013 prices)
`poverty_status`	GLSS (derived)	categorical	Poverty group, cut from welfare at the official lines	Very poor (below 792), Poor (792 to 1,314), Non poor (>=1,314)
`poverty_status_code`	GLSS (derived)	categorical	Numeric code for poverty_status	0 very poor, 1 poor, 2 non poor
`nhis_any_member_registered`	both	binary	Any household member registered/covered by NHIS (binary target)	1 = yes, 0 = no
`nhis_any_missing`	both	binary	Flag: NHIS status missing for some member	1 = some missing, 0 = complete
`totexpeqsc_real`	GLSS	continuous	Total expenditure per adult equivalent	GHS per year (real)
`fdexpeqsc_real`	GLSS	continuous	Food expenditure per adult equivalent	GHS per year (real)
`nfexpeqsc_real`	GLSS	continuous	Non-food expenditure per adult equivalent	GHS per year (real)
`TOTHLTH`	GLSS	continuous	Household health expenditure	GHS per year
`wealth_index`	DHS	categorical	DHS asset wealth quintile (benchmark only, never a predictor)	1 (poorest) ... 5 (richest)
`wealth_score`	DHS	continuous	DHS wealth factor score (benchmark only)	standardized
`survey`	both	id	Which survey the row comes from	GLSS6 or DHS2014
`hh_key`	both	id	Household identifier (links to child file)	text
`household_weight`	both	weight	Sampling weight (use for all rates/means; not for model fitting)	numeric

Child file (DHS only) — outcomes for analysis

One row per under-five child, linked to a DHS household via `hh_key`. These are the outcomes studied *after* enrichment (e.g., stunting by synthetic poverty).

Variable	Source	Type	Description	Values / units
`child_id`	DHS	id	Child identifier (unique within household)	text
`hh_key`	DHS	id	Household link to the household file	text
`child_female`	DHS	binary	Child is female	1 = yes, 0 = no
`child_age_months`	DHS	count	Child age in months	0 to 59
`haz`	DHS	continuous	Height-for-age z-score	z-score
`waz`	DHS	continuous	Weight-for-age z-score	z-score
`whz`	DHS	continuous	Weight-for-height z-score	z-score
`stunted`	DHS	binary	Stunted (haz below -2)	1 = yes, 0 = no
`underweight`	DHS	binary	Underweight (waz below -2)	1 = yes, 0 = no
`wasted`	DHS	binary	Wasted (whz below -2)	1 = yes, 0 = no
`diarrhea_recent`	DHS	binary	Diarrhea in last 2 weeks	1 = yes, 0 = no
`fever_recent`	DHS	binary	Fever in last 2 weeks	1 = yes, 0 = no
`cough_recent`	DHS	binary	Cough in last 2 weeks	1 = yes, 0 = no
`dpt3`	DHS	binary	Received DPT dose 3	1 = yes, 0 = no
`measles`	DHS	binary	Received measles vaccine	1 = yes, 0 = no
`child_weight`	DHS	weight	Child sampling weight	numeric

Note on weights: `household_weight` and `child_weight` are survey sampling weights. Use them for every reported rate or mean. Model fitting in the practical is unweighted (the imputation engine does not take case weights); weights re-enter at the validation and analysis stage.